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| <b>USMAN</b>                                                          | <b>Grade A* Scholar</b>                  |
| Centre Number<br>[ 9 ][ 5 ][ 1 ][ 0 ][ 2 ]                            | Candidate Number<br>[ 0 ][ 0 ][ 5 ][ 5 ] |
| Mentora Academy Internal Candidate Ref:<br><b>MEN-EDX-2026-U4-W05</b> |                                          |

**Pearson Edexcel International Advanced Level**

# Chemistry

**International Advanced Level**

**UNIT 4: Rates, Equilibria and Further Organic Chemistry**

**WEEK 5 ASSESSMENT: TOPIC 12A.3 FEASIBILITY & 12B.1 BORN-HABER CYCLES**

|                   |                                                                                                                                                                   |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Assessment Paper: | <b>WCH14/01 (Week 5 In-Depth Mastery)</b>                                                                                                                         |
| Time Allowed:     | <b>2 hours 30 minutes</b>                                                                                                                                         |
| Total Marks:      | <b>150 Marks</b>                                                                                                                                                  |
| Topics Examined:  | Topic 12A.3: Understanding Entropy Changes (Feasibility & Crossover T)<br>Topic 12B.1: Lattice Energy, LE H and Born-Haber Cycles (Standard Cycles & Definitions) |

## INSTRUCTIONS TO CANDIDATES

- Use black ink or ball-point pen. Fill in the boxes at the top of this page.
- Answer ALL questions in Section A, Section B, and Section C.
- In Section A (Questions 1 to 30), select your answer by placing a clear cross in the box [X].
- In Section B and C, show all working in calculation questions. Include appropriate signs and units where required.
- Questions marked with an asterisk (\*) are those where your quality of written communication and logical sequence of structure will be assessed.
- A Periodic Table and physical constants are printed on the back page of this booklet.

## INFORMATION FOR CANDIDATES

- The total mark for this paper is 150.
- Section A: 30 marks; Section B: 90 marks; Section C: 30 marks.
- Full worked confidential teacher mark scheme is provided at the rear for post-assessment diagnostic.

**SECTION A**

Answer ALL questions. For each question, make a selection by putting a cross in one box [X]. If you change your mind, put a line through the box and then mark your new choice with a cross.

1 Which equation represents the standard lattice energy of sodium chloride, using the formation definition?

- ☐ **A**  $\text{Na(s)} + \frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{NaCl(s)}$
- ☐ **B**  $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) \rightarrow \text{NaCl(s)}$
- ☐ **C**  $\text{NaCl(s)} \rightarrow \text{Na}^+(\text{g}) + \text{Cl}^-(\text{g})$
- ☐ **D**  $\text{Na}^+(\text{aq}) + \text{Cl}^-(\text{aq}) \rightarrow \text{NaCl(s)}$

**(Total for Question 1 = 1 mark)**

2 Which equation defines the first electron affinity of oxygen?

- ☐ **A**  $\text{O(g)} + \text{e}^- \rightarrow \text{O}^-(\text{g})$
- ☐ **B**  $\frac{1}{2} \text{O}_2(\text{g}) + \text{e}^- \rightarrow \text{O}^-(\text{g})$
- ☐ **C**  $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$
- ☐ **D**  $\text{O(s)} + \text{e}^- \rightarrow \text{O}^-(\text{g})$

**(Total for Question 2 = 1 mark)**

3 Why is the second electron affinity of oxygen,  $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$ , endothermic ( $\Delta H > 0$ )?

- ☐ **A** Energy is required to break the covalent  $\text{O}=\text{O}$  bond
- ☐ **B** Work must be done to overcome electrostatic repulsion between the incoming electron and the negative  $\text{O}^-$  ion
- ☐ **C** The  $\text{O}^{2-}$  ion has an unstable noble gas electron configuration
- ☐ **D** The second electron enters a higher principal quantum shell ( $n = 3$ )

**(Total for Question 3 = 1 mark)**

4 Which enthalpy change corresponds to the standard enthalpy of atomisation of chlorine, at  $H^\circ$ ?

- ☐ A  $\text{Cl}_2(\text{g}) \rightarrow 2\text{Cl}(\text{g})$   
☐ B  $\frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$   
☐ C  $\text{Cl}_2(\text{l}) \rightarrow 2\text{Cl}(\text{g})$   
☐ D  $\text{Cl}^-(\text{g}) \rightarrow \text{Cl}(\text{g}) + \text{e}^-$

(Total for Question 4 = 1 mark)

5 Which ionisation enthalpy requires the greatest amount of energy per mole?

- ☐ A First ionisation energy of sodium,  $\text{Na}(\text{g}) \rightarrow \text{Na}^+(\text{g}) + \text{e}^-$   
☐ B First ionisation energy of magnesium,  $\text{Mg}(\text{g}) \rightarrow \text{Mg}^+(\text{g}) + \text{e}^-$   
☐ C Second ionisation energy of magnesium,  $\text{Mg}^+(\text{g}) \rightarrow \text{Mg}^{2+}(\text{g}) + \text{e}^-$   
☐ D Second ionisation energy of sodium,  $\text{Na}^+(\text{g}) \rightarrow \text{Na}^{2+}(\text{g}) + \text{e}^-$

(Total for Question 5 = 1 mark)

6 A reaction has  $H = +90 \text{ kJ mol}^{-1}$  and  $S_{\text{system}} = +250 \text{ J K}^{-1} \text{ mol}^{-1}$ . At what temperature does the reaction become feasible?

- ☐ A Below 360 K  
☐ B Above 360 K  
☐ C Above 2780 K  
☐ D It is feasible at all temperatures

(Total for Question 6 = 1 mark)

7 For a reaction where  $H < 0$  and  $S_{\text{system}} < 0$ , what is the shape of the graph of  $S_{\text{total}}$  against temperature  $T$ ?

- ☐ **A** A horizontal straight line with constant positive slope
- ☐ **B** A curve that starts high and decreases as temperature increases
- ☐ **C** A curve that starts low and increases linearly as temperature increases
- ☐ **D** A vertical line intersecting the temperature axis at 298 K

(Total for Question 7 = 1 mark)

8 Given data:  $\Delta H^\circ_f[\text{NaCl(s)}] = -411 \text{ kJ mol}^{-1}$ ,  $\Delta H^\circ_{\text{at}}[\text{Na}] = +107 \text{ kJ mol}^{-1}$ ,  $1^{\text{st}} \text{ IE}[\text{Na}] = +496 \text{ kJ mol}^{-1}$ ,  $\Delta H^\circ_{\text{at}}[\text{Cl}] = +122 \text{ kJ mol}^{-1}$ ,  $1^{\text{st}} \text{ EA}[\text{Cl}] = -349 \text{ kJ mol}^{-1}$ . What is the lattice energy of formation of  $\text{NaCl(s)}$ ?

- ☐ **A**  $-787 \text{ kJ mol}^{-1}$
- ☐ **B**  $+787 \text{ kJ mol}^{-1}$
- ☐ **C**  $-545 \text{ kJ mol}^{-1}$
- ☐ **D**  $-889 \text{ kJ mol}^{-1}$

(Total for Question 8 = 1 mark)

9 Which of the following compounds would be expected to have the most exothermic lattice energy of formation?

- ☐ **A**  $\text{NaCl}$
- ☐ **B**  $\text{MgCl}_2$
- ☐ **C**  $\text{MgO}$
- ☐ **D**  $\text{BaO}$

(Total for Question 9 = 1 mark)

10 Which enthalpy change is represented by the downward arrow in a standard Born-Haber cycle from gaseous ions to the ionic solid?

- ☐ A Enthalpy of formation
- ☐ B Lattice energy of formation
- ☐ C Enthalpy of atomisation
- ☐ D Enthalpy of hydration

(Total for Question 10 = 1 mark)

11 Consider the hypothetical compound calcium(I) chloride,  $\text{CaCl(s)}$ . Why does  $\text{CaCl(s)}$  spontaneously disproportionate into  $\text{Ca(s)}$  and  $\text{CaCl}_2\text{(s)}$  under standard conditions?

- ☐ A The first ionisation energy of calcium is endothermic
- ☐ B The lattice energy of  $\text{CaCl}_2$  is much more exothermic than that of  $\text{CaCl}$  due to the  $2+$  charge of  $\text{Ca}^{2+}$
- ☐ C Calcium(I) chloride has a covalent molecular structure
- ☐ D Chlorine cannot form negative chloride ions in the presence of  $\text{Ca}^+$

(Total for Question 11 = 1 mark)

12 Which statement regarding the crossover temperature,  $T_{\text{crossover}}$ , is correct?

- ☐ A At  $T = T_{\text{crossover}}$ , the equilibrium constant  $K$  must equal 0
- ☐ B At  $T = T_{\text{crossover}}$ ,  $S_{\text{total}} = 0$  and  $S_{\text{surroundings}} = -S_{\text{system}}$
- ☐ C At  $T = T_{\text{crossover}}$ , the reaction stops occurring at the molecular level
- ☐ D  $T_{\text{crossover}}$  can be calculated for reactions where  $H$  and  $S_{\text{system}}$  have opposite signs

(Total for Question 12 = 1 mark)

13 For the reaction:  $2\text{SO}_2(\text{g}) + \text{O}_2(\text{g}) \rightleftharpoons 2\text{SO}_3(\text{g})$ ,  $H = -197 \text{ kJ mol}^{-1}$  and  $S_{\text{system}} = -188 \text{ J K}^{-1} \text{ mol}^{-1}$ .

Above what temperature does this reaction cease to be spontaneous?

- ☐ A 1048 K
- ☐ B 1.05 K
- ☐ C 955 K
- ☐ D 298 K

(Total for Question 13 = 1 mark)

14 Which reaction is entropy-driven (spontaneous because  $S_{\text{system}}$  is positive and outweighs an unfavourable endothermic  $H$ )?

- ☐ A  $\text{H}_2(\text{g}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow \text{H}_2\text{O}(\text{l})$   $H = -286 \text{ kJ mol}^{-1}$
- ☐ B  $\text{NH}_4\text{NO}_3(\text{s}) + \text{aq} \rightarrow \text{NH}_4^+(\text{aq}) + \text{NO}_3^-(\text{aq})$   $H = +25.7 \text{ kJ mol}^{-1}$
- ☐ C  $\text{CH}_4(\text{g}) + 2\text{O}_2(\text{g}) \rightarrow \text{CO}_2(\text{g}) + 2\text{H}_2\text{O}(\text{l})$   $H = -890 \text{ kJ mol}^{-1}$
- ☐ D  $2\text{Na}(\text{s}) + \text{Cl}_2(\text{g}) \rightarrow 2\text{NaCl}(\text{s})$   $H = -822 \text{ kJ mol}^{-1}$

(Total for Question 14 = 1 mark)

15 Which factor causes barium chloride,  $\text{BaCl}_2$ , to have a less exothermic lattice energy than magnesium chloride,  $\text{MgCl}_2$ ?

- ☐ A Barium has a higher first ionisation energy than magnesium
- ☐ B  $\text{Ba}^{2+}$  has a larger ionic radius than  $\text{Mg}^{2+}$ , resulting in weaker electrostatic attraction to  $\text{Cl}^-$
- ☐ C Barium chloride is more covalent than magnesium chloride
- ☐ D  $\text{Ba}^{2+}$  has a higher charge density than  $\text{Mg}^{2+}$

(Total for Question 15 = 1 mark)

16 The first electron affinity of chlorine is  $-349 \text{ kJ mol}^{-1}$ , whereas that of bromine is  $-325 \text{ kJ mol}^{-1}$ . Why is the electron affinity of chlorine more exothermic?

- ☐ A The incoming electron in chlorine enters a 3p subshell closer to the nucleus than the 4p subshell in bromine
- ☐ B Chlorine has a lower nuclear charge than bromine
- ☐ C Chlorine has greater electron-electron repulsion than fluorine
- ☐ D Bromine is a liquid under standard conditions

**(Total for Question 16 = 1 mark)**

17 When potassium fluoride, KF, is compared to potassium iodide, KI, what is the expected trend in their lattice energies of formation?

- ☐ A KF is more exothermic than KI because  $\text{F}^-$  is smaller than  $\text{I}^-$
- ☐ B KI is more exothermic than KF because iodide has more electrons
- ☐ C Both have identical lattice energies because the cation is  $\text{K}^+$  in both
- ☐ D KI is more exothermic because iodide is more polarisable

**(Total for Question 17 = 1 mark)**

18 A reaction has  $H = +150 \text{ kJ mol}^{-1}$  and  $S_{\text{system}} = -80 \text{ J K}^{-1} \text{ mol}^{-1}$ . Which statement is correct?

- ☐ A The reaction is feasible above 1875 K
- ☐ B The reaction is feasible below 1875 K
- ☐ C The reaction is not feasible at any temperature
- ☐ D The reaction is feasible at all temperatures

**(Total for Question 18 = 1 mark)**

19 In a Born-Haber cycle for magnesium oxide,  $\text{MgO}$ , which of the following represents the correct expression for  $\Delta_f H^\circ[\text{MgO}(\text{s})]$ ?

- ☐ A  $\Delta_f H[\text{Mg}] + 1^{\text{st}} \text{IE}[\text{Mg}] + 2^{\text{nd}} \text{IE}[\text{Mg}] + \Delta_f H[\text{O}] + 1^{\text{st}} \text{EA}[\text{O}] + 2^{\text{nd}} \text{EA}[\text{O}] + \text{LE H}[\text{MgO}]$
- ☐ B  $\Delta_f H[\text{Mg}] + 1^{\text{st}} \text{IE}[\text{Mg}] + \Delta_f H[\text{O}] + 1^{\text{st}} \text{EA}[\text{O}] - \text{LE H}[\text{MgO}]$
- ☐ C  $\text{LE H}[\text{MgO}] - [\Delta_f H[\text{Mg}] + 1^{\text{st}} \text{IE}[\text{Mg}] + 2^{\text{nd}} \text{IE}[\text{Mg}] + \Delta_f H[\text{O}]]$
- ☐ D  $\Delta_f H[\text{Mg}] + 2^{\text{nd}} \text{IE}[\text{Mg}] + \Delta_f H[\text{O}] + 2^{\text{nd}} \text{EA}[\text{O}] + \text{LE H}[\text{MgO}]$

(Total for Question 19 = 1 mark)

20 Why is the standard enthalpy of atomisation of magnesium ( $+148 \text{ kJ mol}^{-1}$ ) greater than that of sodium ( $+107 \text{ kJ mol}^{-1}$ )?

- ☐ A Magnesium atoms are smaller and contribute 2 delocalised electrons per cation to metallic bonding
- ☐ B Magnesium has a higher electronegativity than sodium
- ☐ C Magnesium has a lower second ionisation energy than sodium
- ☐ D Magnesium has a hexagonal close-packed crystal structure

(Total for Question 20 = 1 mark)

21 What happens to the value of  $S_{\text{total}}$  as the temperature of an exothermic reaction approaches absolute zero ( $0 \text{ K}$ )?

- ☐ A  $S_{\text{total}}$  approaches zero
- ☐ B  $S_{\text{total}}$  approaches negative infinity
- ☐ C  $S_{\text{surroundings}}$  approaches positive infinity, making  $S_{\text{total}}$  very large and positive
- ☐ D  $S_{\text{total}}$  becomes equal to  $S_{\text{system}}$

(Total for Question 21 = 1 mark)

22 Given: 1st  $EA[O] = -141 \text{ kJ mol}^{-1}$  and 2nd  $EA[O] = +798 \text{ kJ mol}^{-1}$ . What is the overall enthalpy change for the process:  $O(g) + 2e^- \rightarrow O^{2-}(g)$ ?

- ☐ A  $+657 \text{ kJ mol}^{-1}$
- ☐ B  $-657 \text{ kJ mol}^{-1}$
- ☐ C  $+939 \text{ kJ mol}^{-1}$
- ☐ D  $-939 \text{ kJ mol}^{-1}$

(Total for Question 22 = 1 mark)

23 If a chemical reaction has a positive  $S_{\text{total}}$  at 298 K, which of the following MUST be true?

- ☐ A The reaction must be exothermic ( $H < 0$ )
- ☐ B The reaction must proceed with an increase in entropy of the system ( $S_{\text{system}} > 0$ )
- ☐ C The reaction is thermodynamically feasible at 298 K
- ☐ D The rate of reaction must be extremely rapid at 298 K

(Total for Question 23 = 1 mark)

24 For the thermal decomposition of potassium chlorate:  $2KClO_3(s) \rightarrow 2KCl(s) + 3O_2(g)$ ,  $H = -78 \text{ kJ mol}^{-1}$ . What are the signs of  $S_{\text{system}}$  and  $S_{\text{surroundings}}$  at 298 K?

- ☐ A  $S_{\text{system}}$ : Positive (+) |  $S_{\text{surroundings}}$ : Positive (+)
- ☐ B  $S_{\text{system}}$ : Positive (+) |  $S_{\text{surroundings}}$ : Negative (-)
- ☐ C  $S_{\text{system}}$ : Negative (-) |  $S_{\text{surroundings}}$ : Positive (+)
- ☐ D  $S_{\text{system}}$ : Negative (-) |  $S_{\text{surroundings}}$ : Negative (-)

(Total for Question 24 = 1 mark)

25 Which of the following processes has  $H = 0$  and  $S_{\text{system}} > 0$ , making it entirely entropy-driven at all temperatures?

- ☐ A Evaporation of water
- ☐ B Expansion of an ideal gas into an evacuated flask at constant temperature
- ☐ C Combustion of propane
- ☐ D Dissolution of sodium hydroxide in water

(Total for Question 25 = 1 mark)

26 In the Born-Haber cycle for aluminium oxide,  $\text{Al}_2\text{O}_3(\text{s})$ , how many moles of electrons are involved in the electron affinity step?

- ☐ A 2 moles
- ☐ B 3 moles
- ☐ C 5 moles
- ☐ D 6 moles

(Total for Question 26 = 1 mark)

27 Which of the following compounds has the highest melting temperature, primarily due to its lattice energy?

- ☐ A NaF
- ☐ B CaO
- ☐ C KBr
- ☐ D CsI

(Total for Question 27 = 1 mark)

28 When a reaction with  $H > 0$  and  $S_{\text{system}} > 0$  reaches its crossover temperature  $T_{\text{crossover}}$ , what is the value of the equilibrium constant  $K$ ?

- ☐ A  $K = 0$
- ☐ B  $K = 1$
- ☐ C  $K = \text{infinity}$
- ☐ D  $K = -1$

(Total for Question 28 = 1 mark)

29 Which species requires the enthalpy of atomisation of a diatomic gas to be multiplied by  $1/2$  in its Born-Haber cycle?

- ☐ A Formation of 1 mole of  $\text{NaCl(s)}$  from  $\text{Na(s)}$  and  $\text{Cl}_2(\text{g})$
- ☐ B Formation of 1 mole of  $\text{MgCl}_2(\text{s})$  from  $\text{Mg(s)}$  and  $\text{Cl}_2(\text{g})$
- ☐ C Formation of 1 mole of  $\text{AlCl}_3(\text{s})$  from  $\text{Al(s)}$  and  $\text{Cl}_2(\text{g})$
- ☐ D Formation of 1 mole of  $\text{CCl}_4(\text{l})$  from  $\text{C(s)}$  and  $\text{Cl}_2(\text{g})$

(Total for Question 29 = 1 mark)

30 Why does calcium oxide,  $\text{CaO}$ , adopt an ionic lattice rather than existing as discrete neutral  $\text{CaO}$  molecules?

- ☐ A The oxygen atom has a high electronegativity
- ☐ B The huge lattice energy released when  $\text{Ca}^{2+}$  and  $\text{O}^{2-}$  pack into a crystal lattice outweighs the endothermic cost of forming  $\text{Ca}^{2+}$  and  $\text{O}^{2-}$  ions
- ☐ C The second electron affinity of oxygen is exothermic
- ☐ D Calcium metal easily reacts with atmospheric nitrogen

(Total for Question 30 = 1 mark)

**SECTION B**

Answer ALL questions. Write your answers in the spaces provided.

31 Magnesium chloride,  $\text{MgCl}_2$ , is an ionic compound widely used in the industrial manufacture of magnesium metal by molten electrolysis.

The table below lists thermodynamic data required to construct a Born-Haber cycle for magnesium chloride at 298 K:

| Enthalpy Change                                             | Symbol / Process                                 | Value / $\text{kJ mol}^{-1}$ |
|-------------------------------------------------------------|--------------------------------------------------|------------------------------|
| Standard enthalpy of formation of $\text{MgCl}_2(\text{s})$ | $H_f^\circ[\text{MgCl}_2(\text{s})]$             | -641.3                       |
| Standard enthalpy of atomisation of magnesium               | $H_{\text{at}}^\circ[\text{Mg}(\text{s})]$       | +147.7                       |
| First ionisation energy of magnesium                        | 1st IE $[\text{Mg}]$                             | +737.7                       |
| Second ionisation energy of magnesium                       | 2nd IE $[\text{Mg}]$                             | +1450.7                      |
| Standard enthalpy of atomisation of chlorine                | $H_{\text{at}}^\circ[1/2 \text{Cl}_2(\text{g})]$ | +121.7                       |
| First electron affinity of chlorine                         | 1st EA $[\text{Cl}(\text{g})]$                   | -348.8                       |

(a) Define standard lattice energy (using the lattice formation convention).

(2)

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(b) In the box below, complete the Born-Haber cycle for magnesium chloride by:

- writing chemical species (including state symbols) on each horizontal energy level
- labelling each arrow with the appropriate enthalpy term.

(5)

Enthalpy / Energy

(c) Calculate the standard lattice energy of formation,  $H_{LE}^{\circ}[\text{MgCl}_2(\text{s})]$ , in  $\text{kJ mol}^{-1}$ . Include a sign in your answer.

(4)

$H_{LE}^{\circ}[\text{MgCl}_2(\text{s})] = \dots\dots\dots \text{kJ mol}^{-1}$

(d) Explain why the standard enthalpy of atomisation of chlorine is exactly half the bond dissociation enthalpy of the Cl-Cl bond in  $\text{Cl}_2(\text{g})$ .

(2)

(e) The lattice energy of barium chloride,  $\text{BaCl}_2$ , is  $-2056 \text{ kJ mol}^{-1}$ . Explain, in terms of ionic properties, why the lattice energy of magnesium chloride is significantly more exothermic than that of barium chloride.

(5)

(Total for Question 31 = 18 marks)

32 Calcium oxide,  $\text{CaO(s)}$ , is a basic refractory oxide with a very high melting temperature ( $2613\text{ }^{\circ}\text{C}$ ).

Thermodynamic data for calcium oxide at 298 K are shown below:

| Enthalpy Change                                                        | Value / $\text{kJ mol}^{-1}$ |
|------------------------------------------------------------------------|------------------------------|
| Standard enthalpy of formation of $\text{CaO(s)}$                      | -635.1                       |
| Standard enthalpy of atomisation of calcium, $\text{Ca(s)}$            | +178.2                       |
| First ionisation energy of calcium                                     | +589.8                       |
| Second ionisation energy of calcium                                    | +1145.4                      |
| Standard enthalpy of atomisation of oxygen, $1/2 \text{O}_2(\text{g})$ | +249.2                       |
| First electron affinity of oxygen, $\text{O(g)}$                       | -141.1                       |
| Second electron affinity of oxygen, $\text{O}^-(\text{g})$             | +798.0                       |

(a) Write the chemical equation, including state symbols, that represents the second electron affinity of oxygen.

(2)

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(b) Explain why the first electron affinity of oxygen is exothermic ( $-141.1\text{ kJ mol}^{-1}$ ), whereas the second electron affinity is highly endothermic ( $+798.0\text{ kJ mol}^{-1}$ ).

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(c) Calculate the lattice energy of formation of calcium oxide,  $H_{\text{LE}}^{\circ}[\text{CaO(s)}]$ , in  $\text{kJ mol}^{-1}$ . Include a sign in your answer.

(4)

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$H_{\text{LE}}^{\circ}[\text{CaO(s)}] = \dots\dots\dots \text{kJ mol}^{-1}$

(d) The lattice energy of sodium oxide,  $\text{Na}_2\text{O}$ , is  $-2488 \text{ kJ mol}^{-1}$ , which is considerably less exothermic than that of calcium oxide (approx  $-3400 \text{ kJ mol}^{-1}$ ).  
Given that  $\text{Na}^+$  (0.102 nm) and  $\text{Ca}^{2+}$  (0.100 nm) have almost identical ionic radii, explain why the lattice energy of  $\text{CaO}$  is roughly  $1000 \text{ kJ mol}^{-1}$  more exothermic than that of  $\text{Na}_2\text{O}$ .

**(4)**

(e) Despite the formation of  $\text{O}^{2-}(\text{g})$  ions requiring an overall input of  $+656.9 \text{ kJ mol}^{-1}$  of energy from neutral  $\text{O}(\text{g})$  atoms, ionic metal oxides containing  $\text{O}^{2-}$  ions are universally stable solid compounds. Explain this thermodynamic paradox by considering the energetic contribution of the crystal lattice.

**(5)****(Total for Question 32 = 18 marks)**

33 Sodium hydrogencarbonate (baking soda),  $\text{NaHCO}_3$ , acts as a leavening agent in baking because it decomposes upon heating to release carbon dioxide gas:



Standard molar entropies at 298 K are tabulated below:

| Substance                                   | $\text{NaHCO}_3(\text{s})$ | $\text{Na}_2\text{CO}_3(\text{s})$ | $\text{H}_2\text{O}(\text{g})$ | $\text{CO}_2(\text{g})$ |
|---------------------------------------------|----------------------------|------------------------------------|--------------------------------|-------------------------|
| $S^\circ / \text{J K}^{-1} \text{mol}^{-1}$ | 102.1                      | 136.0                              | 188.7                          | 213.6                   |

(a) Explain why the entropy change of the system,  $S^\circ_{\text{system}}$ , is expected to be large and positive for this reaction.

(2)

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(b) Calculate  $S^\circ_{\text{system}}$  for this reaction in  $\text{J K}^{-1} \text{mol}^{-1}$ .

(3)

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$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(c) Calculate  $S_{\text{surroundings}}$  and  $S_{\text{total}}$  for the reaction at 298 K (25 °C).

Hence explain why baking soda can be stored in an airtight kitchen cupboard for years without decomposing.

(4)

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$S_{\text{total}}(298 \text{ K}) = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(d) Calculate the minimum temperature, in °C, at which the decomposition of sodium hydrogencarbonate becomes thermodynamically feasible.

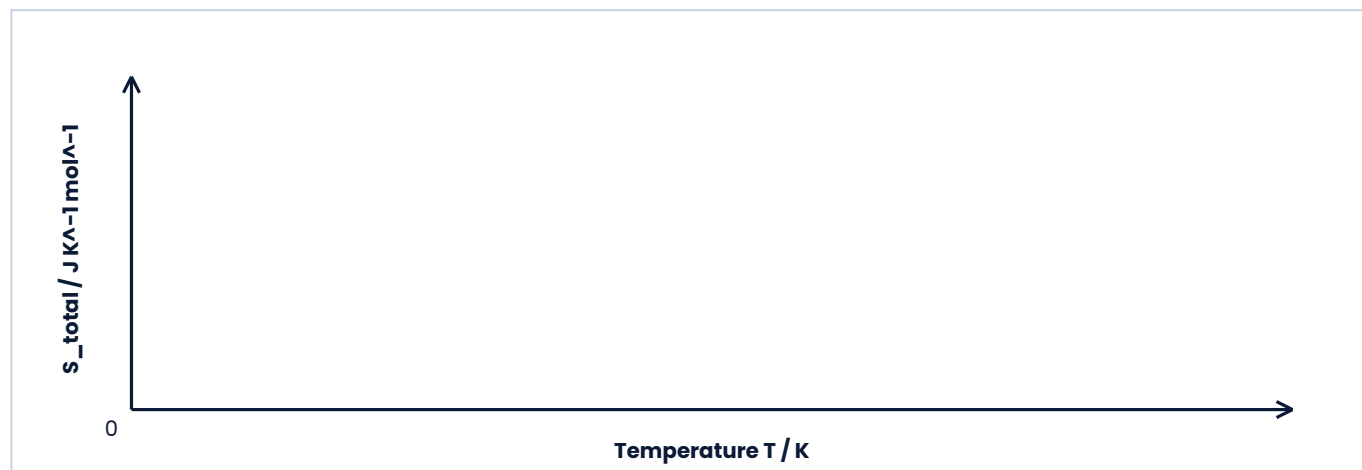
(4)

Minimum temperature = ..... °C

(e) In the space below, sketch a graph of  $S_{\text{total}}$  against temperature  $T$  (from 250 K to 500 K) for this decomposition.

Clearly label the crossover temperature on the temperature axis and the region where the reaction is feasible.

(5)



(Total for Question 33 = 18 marks)

\*34 Magnesium forms the stable divalent ionic halide magnesium chloride,  $\text{MgCl}_2$ .

Neither magnesium(I) chloride,  $\text{MgCl}$ , nor magnesium(III) chloride,  $\text{MgCl}_3$ , can be isolated as stable compounds under standard conditions.

Theoretical and experimental thermodynamic values are tabulated below:

| Compound / Ion                                                                                  | Estimated Lattice Energy / $\text{kJ mol}^{-1}$ | Estimated $H_f^\circ$ / $\text{kJ mol}^{-1}$ |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------|----------------------------------------------|
| $\text{MgCl(s)}$                                                                                | -753                                            | -125                                         |
| $\text{MgCl}_2(\text{s})$                                                                       | -2526                                           | -641                                         |
| $\text{MgCl}_3(\text{s})$                                                                       | -5440                                           | +3920                                        |
| Ionisation Energies of Mg: 1st IE = +738   2nd IE = +1451   3rd IE = +7733 $\text{kJ mol}^{-1}$ |                                                 |                                              |

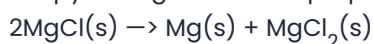
(a) Explain why the estimated lattice energy of  $\text{MgCl}_2$  ( $-2526 \text{ kJ mol}^{-1}$ ) is more than three times as exothermic as that of  $\text{MgCl}$  ( $-753 \text{ kJ mol}^{-1}$ ).

(3)

\*(b) Using the thermodynamic data in the table and principles of Born-Haber cycles, explain why magnesium chloride forms as  $\text{MgCl}_2$  and NOT as  $\text{MgCl}$  or  $\text{MgCl}_3$ .

In your answer, you should include:

- an explanation of why  $\text{MgCl}_3$  is thermodynamically unstable with respect to its elements, referring to the 3rd ionisation energy of magnesium
- an evaluation of why  $\text{MgCl}$  is thermodynamically unstable, including the calculation of the enthalpy change for the disproportionation reaction:



- a deduction of why  $\text{MgCl}_2$  is the uniquely stable product.

(6)

(c) Explain why the third ionisation energy of magnesium ( $+7733 \text{ kJ mol}^{-1}$ ) is so drastically greater than the second ionisation energy ( $+1451 \text{ kJ mol}^{-1}$ ).

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(d) If a chemist attempted to synthesise  $\text{MgCl(s)}$  at low temperatures where disproportionation is kinetically inhibited, predict whether the compound would be diamagnetic or paramagnetic. Justify your answer using electronic configuration.

**(5)**

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**(Total for Question 34 = 18 marks)**

35 In the industrial extraction of iron inside a blast furnace, haematite (iron(III) oxide) is reduced by carbon monoxide:



Thermodynamic data at 298 K are provided in the table below:

| Substance                                   | $\text{Fe}_2\text{O}_3(\text{s})$ | $\text{CO}(\text{g})$ | $\text{Fe}(\text{s})$ | $\text{CO}_2(\text{g})$ |
|---------------------------------------------|-----------------------------------|-----------------------|-----------------------|-------------------------|
| $S^\circ / \text{J K}^{-1} \text{mol}^{-1}$ | 87.4                              | 197.6                 | 27.3                  | 213.6                   |

(a) Calculate the standard entropy change of the system,  $S^\circ_{\text{system}}$ , in  $\text{J K}^{-1} \text{mol}^{-1}$ , for the reduction of iron(III) oxide by carbon monoxide.

(3)

$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(b) Calculate  $S_{\text{surroundings}}$  and  $S_{\text{total}}$  for this reaction at 298 K.

State whether the reaction is thermodynamically feasible at room temperature.

(4)

$S_{\text{total}} (298 \text{ K}) = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(c) At the base of the blast furnace, temperatures exceed 1800 K (approx 1500 °C).

Calculate  $S_{\text{total}}$  for the reduction of iron(III) oxide by carbon monoxide at 1800 K, assuming  $H^\circ$  and  $S^\circ_{\text{system}}$  are independent of temperature.

(3)

$S_{\text{total}}(1800 \text{ K}) = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(d) At these very high temperatures, iron(III) oxide can also be reduced directly by solid coke (carbon):



Given that  $S^\circ[\text{C}(\text{s})] = 5.7 \text{ J K}^{-1} \text{mol}^{-1}$ , calculate  $S^\circ_{\text{system}}$  for this direct reduction.

Hence determine the crossover temperature, in K, above which direct reduction by carbon becomes thermodynamically feasible.

(5)

$T_{\text{crossover}} = \dots\dots\dots \text{K}$

(e) Explain why the direct reduction of iron(III) oxide by carbon has a vastly larger positive  $S^\circ_{\text{system}}$  (+541  $\text{J K}^{-1} \text{mol}^{-1}$ ) than the reduction by carbon monoxide (+15  $\text{J K}^{-1} \text{mol}^{-1}$ ).

(3)

(Total for Question 35 = 18 marks)

## SECTION C

Answer ALL questions. Write your answers in the spaces provided.

36 Hydrated barium chloride,  $\text{BaCl}_2 \cdot 2\text{H}_2\text{O}(\text{s})$ , loses its water of crystallisation upon heating:



Because the direct enthalpy change of dehydration cannot be measured accurately in a school calorimeter, an indirect Hess cycle is constructed using enthalpies of solution:



(a) Construct a Hess's Law cycle linking the reactions above, and calculate the standard enthalpy change of dehydration,  $H^\circ_{\text{dehyd}}$  for:



$H^\circ_{\text{dehyd}} = \dots\dots\dots \text{kJ mol}^{-1}$

(b) Literature values for standard molar entropies at 298 K are:

$S^\circ[\text{BaCl}_2 \cdot 2\text{H}_2\text{O}(\text{s})] = 203.0$  |  $S^\circ[\text{BaCl}_2(\text{s})] = 123.7$  |  $S^\circ[\text{H}_2\text{O}(\text{g})] = 188.7 \text{ J K}^{-1} \text{ mol}^{-1}$ .

Calculate  $S^\circ_{\text{system}}$  for this dehydration in  $\text{J K}^{-1} \text{ mol}^{-1}$ . (3)

$S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{ mol}^{-1}$

(c) Calculate the dehydration temperature (in °C) at which barium chloride dihydrate begins to lose its water of crystallisation spontaneously into the gas phase (where  $S_{\text{total}} \geq 0$ ).

**(4)**

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**Dehydration temperature = ..... °C**

(d) In an industrial drying oven operated at 120 °C, the partial pressure of water vapour is 0.20 atm. Explain how reducing the ambient atmospheric pressure inside a vacuum drying oven lowers the temperature needed to completely dehydrate barium chloride crystals.

**(4)**

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**(Total for Question 36 = 15 marks)**

37 Titanium is a vital aerospace metal. Unlike iron, titanium cannot be extracted from its mineral rutile,  $\text{TiO}_2$ , by direct reduction with carbon because it forms brittle titanium carbide,  $\text{TiC}$ .

Instead, the Kroll process is employed in two distinct stages:



| Species                                     | $\text{TiCl}_4(\text{g})$ | $\text{Mg}(\text{l})$ | $\text{Ti}(\text{s})$ | $\text{MgCl}_2(\text{l})$ |
|---------------------------------------------|---------------------------|-----------------------|-----------------------|---------------------------|
| $H_f^\circ / \text{kJ mol}^{-1}$            | -763.2                    | +8.9                  | 0.0                   | -601.2                    |
| $S^\circ / \text{J K}^{-1} \text{mol}^{-1}$ | 354.9                     | 41.8                  | 30.7                  | 128.4                     |

(a) For Stage 2 of the Kroll process at 1273 K (1000 °C):

(i) Calculate the standard enthalpy change of reaction,  $H^\circ$ , in  $\text{kJ mol}^{-1}$ .

(ii) Calculate the standard entropy change of the system,  $S^\circ_{\text{system}}$ , in  $\text{J K}^{-1} \text{mol}^{-1}$ .

(5)

$H^\circ = \dots\dots\dots \text{kJ mol}^{-1}$        $S^\circ_{\text{system}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(b) Calculate  $S_{\text{surroundings}}$  and  $S_{\text{total}}$  for Stage 2 at 1273 K. Deduce whether this reduction reaction is thermodynamically feasible at 1000 °C.

(4)

$S_{\text{surroundings}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$        $S_{\text{total}} = \dots\dots\dots \text{J K}^{-1} \text{mol}^{-1}$

(c) Explain why Stage 2 must be carried out under an inert atmosphere of pure argon gas rather than in air.

(2)

(d) Titanium is significantly more expensive than iron. Identify two major factors that make the Kroll process an expensive industrial operation, referring to the thermodynamic and operational requirements of the process.

(4)

**(Total for Question 37 = 15 marks)**

**DATA SHEET: PERIODIC TABLE OF THE ELEMENTS & PHYSICAL CONSTANTS**

| Physical Constant               | Symbol     | Numerical Value & Units                                                         |
|---------------------------------|------------|---------------------------------------------------------------------------------|
| Gas constant                    | R          | $8.314 \text{ J mol}^{-1} \text{ K}^{-1}$                                       |
| Avogadro constant               | L or $N_A$ | $6.022 \times 10^{23} \text{ mol}^{-1}$                                         |
| Ionic product of water (298 K)  | $K_w$      | $1.00 \times 10^{-14} \text{ mol}^2 \text{ dm}^{-6}$                            |
| Specific heat capacity of water | c          | $4.18 \text{ J g}^{-1} \text{ K}^{-1} = 4.18 \text{ kJ kg}^{-1} \text{ K}^{-1}$ |
| Standard pressure               | $p^\circ$  | $1.00 \times 10^5 \text{ Pa} = 100 \text{ kPa} = 1.00 \text{ bar}$              |
| Standard temperature            | $T^\circ$  | $298.15 \text{ K} = 25.0 \text{ }^\circ\text{C}$                                |

| Group →  | 1           | 2           | 3           | 4           | 5           | 6           | 7           | 0 / 8       |
|----------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Period 1 | H<br>1.0    |             |             |             |             |             |             | He<br>4.0   |
| Period 2 | Li<br>6.9   | Be<br>9.0   | B<br>10.8   | C<br>12.0   | N<br>14.0   | O<br>16.0   | F<br>19.0   | Ne<br>20.2  |
| Period 3 | Na<br>23.0  | Mg<br>24.3  | Al<br>27.0  | Si<br>28.1  | P<br>31.0   | S<br>32.1   | Cl<br>35.5  | Ar<br>39.9  |
| Period 4 | K<br>39.1   | Ca<br>40.1  | Ga<br>69.7  | Ge<br>72.6  | As<br>74.9  | Se<br>79.0  | Br<br>79.9  | Kr<br>83.8  |
| Period 5 | Rb<br>85.5  | Sr<br>87.6  | In<br>114.8 | Sn<br>118.7 | Sb<br>121.8 | Te<br>127.6 | I<br>126.9  | Xe<br>131.3 |
| Period 6 | Cs<br>132.9 | Ba<br>137.3 | Tl<br>204.4 | Pb<br>207.2 | Bi<br>209.0 | Po<br>[209] | At<br>[210] | Rn<br>[222] |

**CONFIDENTIAL TEACHER MARK SCHEME & EXAMINER TRAP COMMENTARY**

Pearson Edexcel International Advanced Level Chemistry – Unit 4 (WCH14/01) – Week 5 Assessment (150 Marks)

| Q  | Correct Answer | Key Concept & Examiner Guidance                                                                                                                                                                   | Mark |
|----|----------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|
| 1  | B              | Lattice energy of formation is the enthalpy change when 1 mol of ionic solid is formed from gaseous ions: $\text{Na}^+(\text{g}) + \text{Cl}^-(\text{g}) \rightarrow \text{NaCl}(\text{s})$ .     | 1    |
| 2  | A              | 1st electron affinity is the enthalpy change when 1 mol of electrons is added to 1 mol of gaseous atoms: $\text{O}(\text{g}) + \text{e}^- \rightarrow \text{O}^-(\text{g})$ .                     | 1    |
| 3  | B              | Second EA is endothermic due to strong electrostatic repulsion between incoming $\text{e}^-$ and negative $\text{O}^-$ anion.                                                                     | 1    |
| 4  | B              | Enthalpy of atomisation forms 1 mole of gaseous atoms from the element in standard state: $\frac{1}{2} \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$ .                                   | 1    |
| 5  | D              | 2nd IE of sodium removes an electron from a stable, inner noble gas shell ( $2\text{p}^6$ ) with high effective nuclear charge.                                                                   | 1    |
| 6  | B              | Feasible when $S_{\text{total}} > 0 \Rightarrow T > H / S_{\text{sys}} = 90\,000 / 250 = 360\text{ K}$ (above 360 K).                                                                             | 1    |
| 7  | B              | $S_{\text{total}} = S_{\text{sys}} - H/T$ ; as T increases, $-H/T$ decreases, so $S_{\text{total}}$ decreases from high to low.                                                                   | 1    |
| 8  | A              | $\Delta H = \Delta H_f - [\Delta H_f(\text{Na}) + \text{IE}(\text{Na}) + \Delta H_f(\text{Cl}) + \text{EA}(\text{Cl})] = -411 - [107 + 496 + 122 - 349] = -411 - 376 = -787\text{ kJ mol}^{-1}$ . | 1    |
| 9  | C              | $\text{MgO}$ contains small $\text{Mg}^{2+}$ and small $\text{O}^{2-}$ with high charges (+2 and -2). Lattice energy scales with $(q_1 \cdot q_2)/r$ .                                            | 1    |
| 10 | B              | The final arrow from gaseous ions down to the solid crystal lattice represents lattice energy of formation.                                                                                       | 1    |
| 11 | B              | The lattice energy of $\text{MgCl}_2$ ( $-2526\text{ kJ mol}^{-1}$ ) releases massive energy due to 2+ charge, making disproportionation favourable.                                              | 1    |
| 12 | B              | At crossover temperature, $S_{\text{total}} = 0$ , meaning $S_{\text{surroundings}} = -S_{\text{system}}$ .                                                                                       | 1    |
| 13 | A              | $T_{\text{crossover}} = H / S_{\text{sys}} = (-197\,000) / (-188) = 1047.9\text{ K} \approx 1048\text{ K}$ .                                                                                      | 1    |
| 14 | B              | Ammonium nitrate dissolving is endothermic ( $H > 0$ ), yet spontaneous because lattice breakdown increases system entropy immensely.                                                             | 1    |
| 15 | B              | $\text{Ba}^{2+}$ has a much larger ionic radius than $\text{Mg}^{2+}$ , reducing electrostatic attraction to chloride ions.                                                                       | 1    |
| 16 | A              | $\text{Cl}$ is smaller than $\text{Br}$ , so the incoming electron is closer to the nucleus and attracted more strongly.                                                                          | 1    |
| 17 | A              | $\text{F}^-$ is significantly smaller than $\text{I}^-$ , so inter-ionic distance is shorter and lattice energy is more exothermic.                                                               | 1    |
| 18 | C              | $H > 0$ and $S_{\text{sys}} < 0$ : $S_{\text{surr}} < 0$ and $S_{\text{sys}} < 0 \Rightarrow S_{\text{total}}$ is negative at all temperatures (never feasible).                                  | 1    |
| 19 | A              | Standard Hess sum for formation: sum of all upward atomisation/IE/EA steps plus downward lattice energy.                                                                                          | 1    |
| 20 | A              | $\text{Mg}^{2+}$ has higher charge (+2 vs +1) and smaller ionic radius than $\text{Na}^+$ , yielding stronger metallic bonding.                                                                   | 1    |
| 21 | C              | As $T \rightarrow 0\text{ K}$ , $-H/T$ approaches +infinity for exothermic reactions, making $S_{\text{total}}$ enormously positive.                                                              | 1    |
| 22 | A              | Overall EA = 1st EA + 2nd EA = $-141 + 798 = +657\text{ kJ mol}^{-1}$ (net endothermic).                                                                                                          | 1    |
| 23 | C              | $S_{\text{total}} > 0$ is the universal condition for thermodynamic feasibility.                                                                                                                  | 1    |
| 24 | A              | Gas produced (3 mol $\text{O}_2$ from 2 mol solid) $\Rightarrow S_{\text{sys}} > 0$ ; Exothermic ( $H < 0$ ) $\Rightarrow S_{\text{surr}} > 0$ .                                                  | 1    |

|    |   |                                                                                                                                                                                               |   |
|----|---|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|
| 25 | B | Free expansion of ideal gas has $H = 0$ (no intermolecular forces) and $S_{\text{sys}} > 0$ (greater volume/microstates).                                                                     | 1 |
| 26 | D | $\text{Al}_2\text{O}_3$ contains 3 $\text{O}^{2-}$ ions per formula unit; each O atom gains 2 electrons $\Rightarrow$ total 6 moles of electrons.                                             | 1 |
| 27 | B | $\text{CaO}$ has divalent ions ( $\text{Ca}^{2+}$ and $\text{O}^{2-}$ ) with small radii, producing immense lattice energy and very high melting point.                                       | 1 |
| 28 | B | At crossover temperature, $S_{\text{total}} = 0$ ; since $S_{\text{total}} = R \ln K$ , $\ln K = 0 \Rightarrow K = 1$ .                                                                       | 1 |
| 29 | A | $\text{NaCl}$ contains 1 Cl atom per formula unit, so $1/2$ mole of $\text{Cl}_2(\text{g})$ is atomised: $1/2 \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$ .                        | 1 |
| 30 | B | The huge lattice energy of formation of $\text{Ca}^{2+}\text{O}^{2-}$ ( $-3400 \text{ kJ mol}^{-1}$ ) easily pays back the endothermic cost of forming $\text{Ca}^{2+}$ and $\text{O}^{2-}$ . | 1 |

**Question 31: Born-Haber Cycle for Magnesium Chloride (18 Marks)**

(a) [2 Marks]

- The enthalpy change when one mole of an ionic crystalline solid [1]
- is formed from its constituent gaseous ions under standard conditions [1].

(b) [5 Marks]

- Correct energy level sequence:  $\text{Mg}(\text{s}) + \text{Cl}_2(\text{g})$  at datum level,  $\text{Mg}(\text{g}) + \text{Cl}_2(\text{g})$ ,  $\text{Mg}^+(\text{g}) + \text{Cl}_2(\text{g}) + \text{e}^-$ ,  $\text{Mg}^{2+}(\text{g}) + \text{Cl}_2(\text{g}) + 2\text{e}^-$  [2]
- Dissociation/atomisation of chlorine:  $\text{Mg}^{2+}(\text{g}) + 2\text{Cl}(\text{g}) + 2\text{e}^-$  [1]
- Electron affinity step:  $\text{Mg}^{2+}(\text{g}) + 2\text{Cl}^-(\text{g})$  [1]
- Downward arrow to  $\text{MgCl}_2(\text{s})$  labeled  $H_{\text{LE}}^\circ$  and downward arrow from elements to  $\text{MgCl}_2(\text{s})$  labeled  $fH^\circ$  [1].

(c) [4 Marks]

- Hess expression:  $fH^\circ = \text{at}H^\circ(\text{Mg}) + 1\text{st IE}(\text{Mg}) + 2\text{nd IE}(\text{Mg}) + 2*\text{at}H^\circ(\text{Cl}) + 2*1\text{st EA}(\text{Cl}) + H_{\text{LE}}^\circ$  [1]
- $-641.3 = +147.7 + 737.7 + 1450.7 + 2(121.7) + 2(-348.8) + H_{\text{LE}}^\circ$  [1]
- $-641.3 = +2336.1 + 243.4 - 697.6 + H_{\text{LE}}^\circ = +1881.9 + H_{\text{LE}}^\circ$  [1]
- $H_{\text{LE}}^\circ = -641.3 - 1881.9 = -2523.2 \text{ kJ mol}^{-1}$  (accept  $-2523$  to  $-2526 \text{ kJ mol}^{-1}$ ) [1].

Examiner Trap: Omitting the factor of 2 for chlorine atomisation or electron affinity loses 2 marks!

(d) [2 Marks]

- Bond enthalpy represents breaking 1 mole of Cl-Cl bonds to form 2 moles of Cl atoms:  $\text{Cl}_2(\text{g}) \rightarrow 2\text{Cl}(\text{g})$  [1]
- Standard enthalpy of atomisation is defined as forming 1 mole of gaseous Cl atoms:  $1/2 \text{Cl}_2(\text{g}) \rightarrow \text{Cl}(\text{g})$  [1].

(e) [5 Marks]

- $\text{Mg}^{2+}$  has a much smaller ionic radius than  $\text{Ba}^{2+}$  (0.072 nm vs 0.135 nm) [1]
- Both cations possess the same +2 ionic charge [1]
- Therefore  $\text{Mg}^{2+}$  has a much higher charge density than  $\text{Ba}^{2+}$  [1]
- The electrostatic attraction between  $\text{Mg}^{2+}$  and  $\text{Cl}^-$  ions is much stronger / shorter inter-ionic distance [1]
- Since lattice energy is proportional to  $(q_1*q_2)/r$ ,  $\text{MgCl}_2$  has a substantially more exothermic lattice energy [1].

**Question 32: Calcium Oxide Born-Haber Cycle & 2nd Electron Affinity (18 Marks)**

(a) [2 Marks]

- $\text{O}^-(\text{g}) + \text{e}^- \rightarrow \text{O}^{2-}(\text{g})$  [1]
- Correct state symbols (g) on both oxygen species [1].

(b) [3 Marks]

- 1st EA is exothermic because the incoming electron is attracted to the positively charged nucleus of the neutral oxygen atom [1]
- 2nd EA involves adding a negative electron to an already negatively charged  $\text{O}^-$  anion [1]
- Strong electrostatic repulsion between like negative charges requires large energy input to force the electron into the orbital [1].

(c) [4 Marks]

- $fH^\circ = \text{at}H(\text{Ca}) + 1\text{st IE}(\text{Ca}) + 2\text{nd IE}(\text{Ca}) + \text{at}H(\text{O}) + 1\text{st EA}(\text{O}) + 2\text{nd EA}(\text{O}) + H_{\text{LE}}^\circ$  [1]
- $-635.1 = +178.2 + 589.8 + 1145.4 + 249.2 + (-141.1) + (+798.0) + H_{\text{LE}}^\circ$  [1]
- $-635.1 = +2819.5 + H_{\text{LE}}^\circ$  [1]

•  $H_{LE}^{\circ} = -635.1 - 2819.5 = -3454.6 \text{ kJ mol}^{-1}$  (accept  $-3454$  to  $-3455 \text{ kJ mol}^{-1}$ ) [1].

(d) [4 Marks]

- $\text{Ca}^{2+}$  has a +2 charge whereas  $\text{Na}^{+}$  has a +1 charge [1]
- Both have almost identical ionic radii ( $\sim 0.10 \text{ nm}$ ), so  $\text{Ca}^{2+}$  has double the charge density of  $\text{Na}^{+}$  [1]
- Electrostatic force / lattice energy is proportional to  $(q_{\text{cation}} \times q_{\text{anion}})$  [1]
- In  $\text{CaO}$ , charges are  $(+2)(-2) = -4$ , whereas in  $\text{Na}_2\text{O}$  charges are  $(+1)(-2) = -2$ , making  $\text{CaO}$  lattice energy  $\sim 1000 \text{ kJ mol}^{-1}$  more exothermic [1].

(e) [5 Marks]

- Forming  $\text{O}_2(\text{g})$  from  $\text{O}(\text{g})$  requires net endothermic energy ( $-141.1 + 798.0 = +656.9 \text{ kJ mol}^{-1}$ ) [1]
- Removing 2 electrons from  $\text{Ca}$  requires  $+1735.2 \text{ kJ mol}^{-1}$  of ionisation energy [1]
- Total endothermic hurdle to create  $\text{Ca}^{2+}(\text{g})$  and  $\text{O}_2^{2-}(\text{g})$  is over  $+2800 \text{ kJ mol}^{-1}$  [1]
- However, the condensation of  $\text{Ca}^{2+}$  and  $\text{O}_2^{2-}$  into a solid crystal lattice releases  $\sim 3455 \text{ kJ mol}^{-1}$  of lattice energy [1]
- The immense lattice energy easily exceeds the endothermic barrier, making overall formation  $\Delta H^{\circ}$  strongly exothermic ( $-635 \text{ kJ mol}^{-1}$ ) [1].

### Question 33: Baking Soda Decomposition & Crossover Feasibility (18 Marks)

(a) [2 Marks]

- 2 moles of solid reactant produce 1 mole of solid and 2 moles of gas ( $\text{H}_2\text{O}(\text{g}) + \text{CO}_2(\text{g})$ ) [1]
- Gaseous particles possess much greater disorder / translational freedom / dispersion of energy quanta [1].

(b) [3 Marks]

- $S^{\circ}_{\text{system}} = [S^{\circ}(\text{Na}_2\text{CO}_3) + S^{\circ}(\text{H}_2\text{O}) + S^{\circ}(\text{CO}_2)] - 2 \times S^{\circ}(\text{NaHCO}_3)$  [1]
- $= [136.0 + 188.7 + 213.6] - 2(102.1) = 538.3 - 204.2$  [1]
- $= +334.1 \text{ J K}^{-1} \text{ mol}^{-1}$  [1].

(c) [4 Marks]

- $S_{\text{surr}} = -H / T = -(+135\,600 \text{ J mol}^{-1}) / 298 \text{ K} = -455.03 \text{ J K}^{-1} \text{ mol}^{-1}$  [1]
- $S_{\text{total}} = +334.1 + (-455.03) = -120.93 \text{ J K}^{-1} \text{ mol}^{-1}$  [2]
- Because  $S_{\text{total}}$  is negative at 298 K, decomposition is thermodynamically non-feasible / cannot occur spontaneously at room temperature [1].

(d) [4 Marks]

- Reaction becomes feasible when  $S_{\text{total}} \geq 0 \Rightarrow T_{\text{min}} = H / S_{\text{system}}$  [1]
- $T_{\text{min}} = 135\,600 \text{ J mol}^{-1} / 334.1 \text{ J K}^{-1} \text{ mol}^{-1} = 405.87 \text{ K}$  [2]
- In  $^{\circ}\text{C}$ :  $405.87 - 273.15 = 132.7^{\circ}\text{C} \approx 133^{\circ}\text{C}$  [1].

(e) [5 Marks]

- Graph axes: y-axis labeled  $S_{\text{total}}$ , x-axis labeled  $T / \text{K}$  [1]
- Curve shape: Starts negative at low  $T$ , rises non-linearly (or asymptotically) as  $T$  increases [1]
- Intersection with T-axis at  $T = 406 \text{ K}$  (labeled  $T_{\text{crossover}}$ ) [1]
- Region above 406 K clearly labeled 'Feasible ( $S_{\text{total}} > 0$ )' [1]
- Region below 406 K clearly labeled 'Non-feasible ( $S_{\text{total}} < 0$ )' [1].

### Question 34: \*Level of Response — $\text{MgCl}$ vs $\text{MgCl}_2$ vs $\text{MgCl}_3$ Stability (18 Marks)

(a) [3 Marks]

- $\text{Mg}^{2+}$  has twice the charge of  $\text{Mg}^{+}$  (+2 vs +1) [1]
- $\text{Mg}^{2+}$  has a smaller ionic radius than  $\text{Mg}^{+}$  because the same nuclear charge pulls fewer electrons [1]
- Lattice energy is directly proportional to charge product and inversely to radius, so  $(2^+)/r_{\text{small}} \gg (1^+)/r_{\text{large}}$  [1].

\*(b) [6 Marks — Level of Response]

- Level 3 (5–6 marks): Evaluates all three compounds with thorough thermodynamic reasoning. Calculates disproportionation enthalpy:  $H = \Delta H(\text{MgCl}_2) - 2\Delta H(\text{MgCl}) = -641 - 2(-125) = -391 \text{ kJ mol}^{-1}$  (highly exothermic, so  $\text{MgCl}$  spontaneously disproportionates). Explains that  $\text{MgCl}_3$  requires removing a 3rd electron from the 2p core shell ( $+7733 \text{ kJ mol}^{-1}$ ), which cannot be compensated by lattice energy ( $-5440 \text{ kJ mol}^{-1}$ ), resulting in massive positive  $\Delta H$  ( $+3920 \text{ kJ mol}^{-1}$ ). Concludes  $\text{MgCl}_2$  is uniquely stable.
- Level 2 (3–4 marks): Correctly calculates disproportionation enthalpy with minor slip; explains 3rd IE barrier for  $\text{MgCl}_3$ ; basic comparison.
- Level 1 (1–2 marks): Mentions octet rule or ion charges without quantitative thermodynamic analysis.

(c) [4 Marks]

- 1st and 2nd IEs remove outer 3s valence electrons [1]
- 3rd IE removes an electron from an inner complete 2p subshell (noble gas neon core) [1]
- The 2p electron is much closer to the nucleus (lower principal quantum number  $n = 2$ ) [1]
- Experiences much less shielding from inner shells, so attraction by 12 protons is immense [1].

(d) [5 Marks]

- Electronic configuration of Mg atom:  $1s^2 2s^2 2p^6 3s^2$  [1]
- Electronic configuration of  $Mg^+$  ion in  $MgCl$ :  $1s^2 2s^2 2p^6 3s^1$  [1]
- $Mg^+$  possesses one unpaired electron in the 3s orbital [1]
- Substances with unpaired electrons are paramagnetic (attracted into magnetic fields) [1]
- In contrast,  $MgCl_2$  contains  $Mg^{2+}$  ( $1s^2 2s^2 2p^6$ ) with all paired electrons, which is diamagnetic [1].

**Question 35: Blast Furnace Thermodynamics: Reduction of Iron Ore (18 Marks)**

(a) [3 Marks]

- $S^\circ_{\text{system}} = [2^\circ S^\circ(\text{Fe}) + 3^\circ S^\circ(\text{CO}_2)] - [^\circ S^\circ(\text{Fe}_2\text{O}_3) + 3^\circ S^\circ(\text{CO})]$  [1]
- $= [2(27.3) + 3(213.6)] - [87.4 + 3(197.6)] = [54.6 + 640.8] - [87.4 + 592.8] = 695.4 - 680.2$  [1]
- $= +15.2 \text{ J K}^{-1} \text{ mol}^{-1}$  [1].

(b) [4 Marks]

- $S_{\text{surr}} = -(-24\,800 \text{ J mol}^{-1}) / 298 \text{ K} = +83.22 \text{ J K}^{-1} \text{ mol}^{-1}$  [2]
- $S_{\text{total}} = +15.2 + 83.22 = +98.42 \text{ J K}^{-1} \text{ mol}^{-1}$  [1]
- Since  $S_{\text{total}} > 0$ , the reaction is thermodynamically feasible at 298 K [1].

(c) [3 Marks]

- At 1800 K:  $S_{\text{surr}} = -(-24\,800) / 1800 = +13.78 \text{ J K}^{-1} \text{ mol}^{-1}$  [1]
- $S_{\text{total}} = +15.2 + 13.78 = +28.98 \text{ J K}^{-1} \text{ mol}^{-1}$  [1]
- Feasible at 1800 K, but  $S_{\text{total}}$  is smaller than at 298 K because reaction is exothermic [1].

(d) [5 Marks]

- $S^\circ_{\text{system}} = [2^\circ S^\circ(\text{Fe}) + 3^\circ S^\circ(\text{CO})] - [^\circ S^\circ(\text{Fe}_2\text{O}_3) + 3^\circ S^\circ(\text{C})]$  [1]
- $= [2(27.3) + 3(197.6)] - [87.4 + 3(5.7)] = [54.6 + 592.8] - [87.4 + 17.1] = 647.4 - 104.5$  [1]
- $= +542.9 \text{ J K}^{-1} \text{ mol}^{-1}$  [1]
- $T_{\text{crossover}} = H / S_{\text{system}} = 490\,700 \text{ J mol}^{-1} / 542.9 \text{ J K}^{-1} \text{ mol}^{-1} = 903.8 \text{ K}$  (approx 631 °C) [2].

(e) [3 Marks]

- Reduction by CO: 3 moles of gas (CO) produce 3 moles of gas ( $\text{CO}_2$ ), so  $n_{\text{gas}} = 0$  (small  $S_{\text{sys}}$ ) [1]
- Direct reduction by C: 0 moles of gas produce 3 moles of gas (CO) from solid reactants ( $n_{\text{gas}} = +3$ ) [1]
- Creating 3 moles of gas from purely solid reactants creates an enormous increase in disorder / microstates [1].

**Question 36: Barium Chloride Dehydration Calorimetry (15 Marks)**

(a) [4 Marks]

- By Hess's Law:  $H^\circ_{\text{dehyd}} = H_2 - H_1 + 2^\circ H_{\text{vap}}$  [1]
- $= (+8.8) - (-12.6) + 2(+40.7)$  [2]
- $= +21.4 + 81.4 = +102.8 \text{ kJ mol}^{-1}$  [1].

(b) [3 Marks]

- $S^\circ_{\text{system}} = [^\circ S^\circ(\text{BaCl}_2) + 2^\circ S^\circ(\text{H}_2\text{O(g)})] - ^\circ S^\circ(\text{BaCl}_2 \cdot 2\text{H}_2\text{O})$  [1]
- $= [123.7 + 2(188.7)] - 203.0 = [123.7 + 377.4] - 203.0 = 501.1 - 203.0$  [1]
- $= +298.1 \text{ J K}^{-1} \text{ mol}^{-1}$  [1].

(c) [4 Marks]

- Dehydration occurs when  $S_{\text{total}} \geq 0 \Rightarrow T_{\text{min}} = H^\circ_{\text{dehyd}} / S^\circ_{\text{system}}$  [1]
- $T_{\text{min}} = 102\,800 \text{ J mol}^{-1} / 298.1 \text{ J K}^{-1} \text{ mol}^{-1} = 344.85 \text{ K}$  [2]
- In °C:  $344.85 - 273.15 = 71.7 \text{ °C} \approx 72 \text{ °C}$  [1].

(d) [4 Marks]

- Dehydration produces 2 moles of water gas:  $\text{BaCl}_2 \cdot 2\text{H}_2\text{O(s)} \rightleftharpoons \text{BaCl}_2\text{(s)} + 2\text{H}_2\text{O(g)}$  [1]
- Lowering pressure removes water vapour from above the solid / decreases partial pressure of  $\text{H}_2\text{O(g)}$  [1]

- According to Le Chatelier's principle, equilibrium shifts to the side with more moles of gas (to the right) [1]
- Thermodynamically, water gas at low partial pressure has higher molar entropy, increasing  $S_{\text{system}}$  and lowering crossover T [1].

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**Question 37: The Kroll Process for Titanium Extraction (15 Marks)**

(a) [5 Marks]

- (i)  $H^\circ = [fH(\text{Ti}) + 2*fH(\text{MgCl}_2)] - [fH(\text{TiCl}_4) + 2*fH(\text{Mg})]$  [1]
- $= [0 + 2(-601.2)] - [-763.2 + 2(8.9)] = -1202.4 - [-763.2 + 17.8] = -1202.4 - (-745.4)$  [1]
- $= -457.0 \text{ kJ mol}^{-1}$  [1]
- (ii)  $S^\circ_{\text{system}} = [S^\circ(\text{Ti}) + 2*S^\circ(\text{MgCl}_2)] - [S^\circ(\text{TiCl}_4) + 2*S^\circ(\text{Mg})] = [30.7 + 2(128.4)] - [354.9 + 2(41.8)]$  [1]
- $= [30.7 + 256.8] - [354.9 + 83.6] = 287.5 - 438.5 = -151.0 \text{ J K}^{-1} \text{ mol}^{-1}$  [1].

(b) [4 Marks]

- At 1273 K:  $S_{\text{surr}} = -(-457\,000 \text{ J mol}^{-1}) / 1273 \text{ K} = +359.0 \text{ J K}^{-1} \text{ mol}^{-1}$  [2]
- $S_{\text{total}} = -151.0 + 359.0 = +208.0 \text{ J K}^{-1} \text{ mol}^{-1}$  [1]
- Since  $S_{\text{total}} > 0$ , the reaction IS thermodynamically feasible at 1000 °C [1].

(c) [2 Marks]

- Titanium and molten magnesium react vigorously with oxygen and nitrogen in air at 1000 °C to form oxides and nitrides [1]
- This would ruin the purity and mechanical properties of the titanium metal [1].

(d) [4 Marks]

- Expensive reagents: Magnesium metal is itself expensive to produce via electrolysis [2]
  - Batch process & high energy costs: Operating at 1000 °C for days under pure argon is a slow batch process requiring immense electrical heating and argon gas [2].
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